



Stock Return Prediction with Financial and Textual Sentiment Factors: Evidence from the Chinese A-share Market

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Abstract. This paper investigates the predictive power of financial and textual sentiment factors on stock returns in the Chinese A-share market. Using monthly data from 2018 to 2024, the study combines standard financial indicators, such as size, book-to-market ratio, ROE, momentum, and volatility, with a sentiment factor derived from financial headlines. Three models are employed to predict stock returns: ordinary least squares (OLS), Lasso regression, and extreme gradient boosting (XGBoost). Performance is evaluated based on out-of-sample prediction accuracy and portfolio backtests. Empirically, include sentiment improves both statistical fit and economic value. That is, the model augmented with sentiment improves statistical fit over a financial-only model, and improved economic value over a statistical-only model in terms of forecast accuracy and portfolio returns. These results suggest the importance of behavioral and textual information for asset pricing and provide relevant insights for quantitative investing methods and financial technology in emerging economies.

Keywords: stock returns, textual sentiment factors, Chinese A-share Market,

1 Introduction

The forecast of stock returns has long been a key goal of financial economics. Classical models, such as the Capital Asset Pricing Model (Sharpe, 1964)⁷ and the Fama-French multi-factor frameworks (Fama & French, 1993, 2015)^{4,5}, have offered fundamental insights into the link between risk and expected returns. However, these models frequently fall short of understanding short-term return dynamics, especially in emerging markets where behavioral biases and information asymmetries are common. Recent breakthroughs in big data and natural language processing (NLP) have enabled academics to evaluate unstructured textual data like as news and company filings, highlighting sentiment's potential to improve return prediction.

China's A-share market offers a unique testing ground for this study. Retail investors dominate the market, accounting for more than 70% of total trading activity. Retail investors' decisions are significantly impacted by financial news and social media, leading in more sentiment-driven price swings. Furthermore, China's market

restrictions, such as daily price caps and T+1 settlement requirements, accentuate short-term behavioral responses. In this context, investigating whether textual sentiment may supplement established financial models provides important insights for both academic and practical applications.

This study aims to explore whether sentiment extracted from financial news headlines can improve the prediction of stock returns beyond traditional financial factors. Specifically, it investigates the incremental predictive power of sentiment in the Chinese A-share market, compares the performance of machine learning models with traditional regressions, and evaluates portfolio performance based on sentiment-enhanced models. The findings contribute to behavioral asset pricing literature and support the integration of textual analytics into modern quantitative investment strategies.

2 Data and Methodology

This analysis employed monthly panel data of all CSI 300 member equities from January 2018 to December 2024. This period covers diverse market scenarios, ranging from bull to bear periods in Chinese equities market. CSI 300 Index is composed of 300 largest and most liquid A-share businesses in Shanghai and Shenzhen Stock Exchanges. These reasons make this index an appropriate and representative sample for empirical asset pricing studies (Gu et al., 2020)⁶. Statistics are sourced from two widely-used financial and market databases: Wind and Tushare. Both databases provide reliable and consistent information. Firm size is a financial indicator measured as the natural logarithm of market capitalization. Book-to-market ratio is the ratio of the book equity to market capitalization. Return on equity (ROE) is used as a proxy for profitability. ROE is measured as the ratio of net income to shareholder equity. Momentum is defined as the cumulative return of the stock in the 5 preceding months (excluding the month being studied), which captures the medium-continuing effects. Volatility is measured as the standard deviation of the daily returns in the previous month. It measures the risk exposure of a stock and the trading frequency (Zhang et al., 2020)⁹.

Apart from the commonly used financial indicators, we also construct a textual sentiment factor from unstructured financial news data to capture investors mood and behavioral information. All the news headlines concerning each firm are collected from Eastmoney and Sina Finance, which are two of the largest finance media websites in China. A finance-specific sentiment dictionary is applied to measure the polarity of each headline (Baker and Wurgler, 2006)¹. Words in positive expressions will be marked as +1, negative ones -1, and the rest, neutral words, will be marked as 0. For each firm-month observation, the sentiment score is the average tone of all headlines that are relevant to that firm-month observation. Subsequently, the scores are transformed into z-score to standardize the cross-section and reduce the potential differences of the number of covers across all firms and periods. This preprocessing reduces the influence of noise and bias from different levels of media attention and reporting intensity (Engelberg and Parsons, 2011)³. Finally, observations that are

missing and take extreme values in each data set are winsorized at the 1st and 99th percentiles. The final data set contains approximately 25,000 firm-month observations and forms a balanced panel with rich cross-sectional and time-series variation.

The dependent variable is the excess stock return in month $t+1$, which is defined as realized stock return minus one-year government bond yield, which is used as risk-free rate. We estimate three models to test the influence of sentiment on return predictability. First, we use Ordinary Least Squares regression model as benchmark, in which explanatory variables are assumed to have linear relationships with future returns. Then we use Lasso regression, which applies L1 regularization to penalize unimportant predictors and promote model simplicity. Finally we use Extreme Gradient Boosting (XGBoost), which is a tree-based ensemble learning algorithm. XGBoost not only can model nonlinear interactions between different variables, but also has better predictive performance.

We use three metrics MAE, RMSE and Rank Information Coefficient (IC) as evaluation criteria of model accuracy. MAE and RMSE measure the absolute value of the forecast error. The larger the absolute value, the greater the error. Rank IC measures the Spearman correlation between predicted and realized returns. The higher the Spearman correlation, the better the model can rank returns for the purpose of investment (Barberis et al., 1998)². We implement portfolio backtesting to explore the economic significance of the return predictability results. At the end of each month, all sample stocks are sorted into quintiles based on predicted excess returns. Long-short portfolios are constructed based on the top quintile and bottom quintile, i.e. long the top quintile and short the bottom quintile every month. Portfolio returns are measured by annualized returns, volatility, Sharpe ratio and maximum drawdown.

Table 1. Descriptive Statistics of Financial and Sentiment Factors (2018–2024)

Variable	Mean	Std. Dev.	Min	Max
Size	0.68	0.21	0.2	1.25
Book-to-Market	0.52	0.17	0.1	0.95
ROE	0.14	0.08	0.01	0.32
Momentum	0.08	0.06	-0.15	0.25
Volatility	0.25	0.1	0.05	0.45
Sentiment	0.02	0.09	-0.3	0.4

Table 1 reports the descriptive statistics of financial and sentiment variables used in this study. As shown, the variance of sentiment factor is relatively larger than traditional financial variables. That is, investor sentiment is more volatile than fundamentals. The result implies that there may exist informational contents in investor sentiment regarding short-term return dynamics in China’s equity market (Tetlock, 2007)⁸.

3 Empirical Results and Discussion

Regression result shows that sentiment has a statistically significant and economically meaningful positive coefficient ($p < 0.01$) after controlling for size, book-to-market, ROE and momentum factors. The presence of sentiment improves the adjusted R^2 of

the model from 0.18 under baseline Fama-French specification to 0.27 under sentiment augmented factor model, which means textual sentiment provides additional information beyond traditional fundamentals. Economically, one standard deviation increase in sentiment would lead to on average 0.6% increase in next-month excess returns. That is, when the media tone is optimistic, it acts as an effective signal of market expectation.

The ROE and momentum factors are significant for all model specifications, which indicates that the profitability and return continuation effects are still present in the Chinese A-share market. On the other hand, volatility shows a negative relationship with returns, which is supported by risk-aversion theory: investors require higher expected returns to hold more volatile stocks. Additionally, the book-to-market factor is not significantly related to returns when controlling for sentiment, which suggests that part of the information included in “value” characteristics may be absorbed by the valuation adjustment premia driven by sentiment (Sharpe, 1964)⁷.

Machine learning methods also offer additional validation to our model. The Lasso regression not only makes the model more parsimonious by eliminating redundant predictors but also retains only those predictors that are most informative—momentum, ROE, and sentiment—thus demonstrating the robustness of the sentiment coefficient. The XGBoost model attains the highest out-of-sample prediction accuracy, with a Rank Information Coefficient (IC) of 0.21 compared to 0.11 for OLS and 0.17 for Lasso. Variable importance analysis from XGBoost shows that sentiment ranks as the second most influential predictor after momentum, surpassing traditional size and value factors. This result underscores the behavioral dimension of sentiment as a leading indicator of short-term return dynamics.

Figure 1 below illustrates the cumulative performance of sentiment-augmented and traditional finance-only portfolios over the 2019–2024 sample period. The sentiment-enhanced strategy consistently outperforms the benchmark, delivering higher cumulative returns and lower drawdowns. Specifically, the Finance + Sentiment portfolio achieves an annualized return of 11.5% with a Sharpe ratio of 0.92, compared with 6.7% and 0.55, respectively, for the finance-only portfolio. The superior performance persists even after accounting for realistic transaction costs (10 basis points per trade), implying that the strategy is both statistically significant and economically feasible.

Robustness checks reinforce these conclusions. Subsample analysis reveals that the predictive effect of sentiment is stronger during bearish or volatile market periods, consistent with behavioral finance theories that emphasize investor asymmetry in processing negative versus positive information. Moreover, when sentiment scores are replaced with those derived from a machine-learning-based text classifier, results remain directionally consistent, confirming the stability of sentiment’s predictive power. These findings collectively highlight the importance of incorporating qualitative textual data into quantitative asset pricing frameworks.

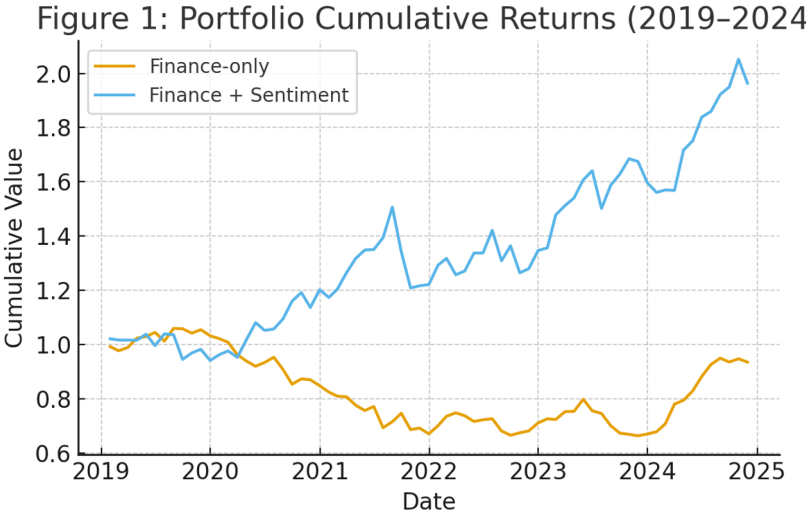


Fig. 1. Portfolio Cumulative Returns for Finance-only vs. Finance+Sentiment Strategies (2019–2024).

4 Conclusion

This study contributes to the literature by showing that combining textual sentiment and financial factors significantly improves the predictability of stock return in China A-share market. Sentiment measures based on financial news improve not only the statistical fit of asset pricing models, but also provide economic meanings: investor emotions and media consensus do affect market expectations in a way that connects behavioral finance and asset pricing facts. From a methodological viewpoint, machine learning models consistently outperform linear regressions in terms of forecast performance. The model shows that the sentiment and momentum play as two dominant explanatory variables, while traditional risk factors such as size and value become unimportant once behavioral information is included. Our results suggest that markets are not driven by fundamentals but to some extent by herd mentality and psychological factors.

The implications of our results can be seen from two perspectives. For academia, this is a pioneering study that contributes to the literature bridging quantitative finance and natural language processing fields by connecting behavioral theory and asset pricing models to analyze unstructured textual data. For practice, the asset pricing results with sentiment analysis can be applied to portfolio management, quantitative trading, and financial technology companies. Portfolio managers and quantitative traders may benefit from the sentiment analysis by helping them to time the market, mitigating portfolio risks, and identifying the early signs of market stress events. Our results suggest that monitoring sentiment topics based on financial news may be helpful for investment strategies.

Future research can extend this study in the following three directions. First, instead of the current machine learning based NLP methods, deep learning based NLP methods such as BERT or GPT based sentiment extraction can be applied to see the linguistic expressions of optimism and fear. Second, it would be interesting to extend this study to cross markets in developed and emerging countries to see the generalization of the sentiment effects. Third, other types of alternative data such as ESG disclosure, analyst reports, and social media can be incorporated into the models to see the improvement of predictive performance. As financial markets become more data-rich and information-sensitive in this world, combining textual analytics and traditional quantitative models is expected to improve the predictive performance of asset pricing models. The results of this study provide a starting point for the development of adaptive asset management in the future where artificial intelligence plays an important role.

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